



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Homework — Series and Parallel Circuits

Every question here can be answered with $V = IR$ and the rules you measured in class. Show every step: an answer with no working earns almost nothing in the exam, and I cannot see where you went wrong.

1) **State** the two rules for a series circuit — one about current, one about potential difference — and the equation for the total resistance of two resistors in series. [3]

The current is the same at every point in the loop (1). The supply pd is shared between the components (1). $R_{total} = R_1 + R_2$ (1).

2) **Calculate** the total resistance, and then the current, in a series circuit where a $4\ \Omega$ resistor and a $2\ \Omega$ resistor are connected to a 12 V supply. [3]

$R_{total} = 4 + 2 = 6\ \Omega$ (1). $I = V/R = 12/6$ (1) = 2.0 A (1).

3) **Calculate** the potential difference across *each* resistor in the circuit from question 2, and show that your two answers are consistent with the supply. [3]

$V = IR$ per component: $V_1 = 2 \times 4 = 8\ \text{V}$ (1); $V_2 = 2 \times 2 = 4\ \text{V}$ (1); $8 + 4 = 12\ \text{V}$, the supply pd — the series pd rule holds (1).

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4) **Calculate** the current in each branch, and the total current from the supply, when a 3Ω resistor and a 6Ω resistor are connected *in parallel* across a 6 V supply. [4]

Parallel: Each branch gets the *full* supply pd — so use $V = IR$ on one branch at a time. You are *not* being asked for a total resistance: Combined Science never asks you to calculate that for parallel resistors.

Each branch has 6 V across it (1). $I_1 = 6/3 = 2.0\text{ A}$ (1); $I_2 = 6/6 = 1.0\text{ A}$ (1). Total from the supply = $2.0 + 1.0 = 3.0\text{ A}$ (1).

5) **Explain** why adding a resistor *in series* increases the total resistance of a circuit, but adding a resistor *in parallel* decreases it. No calculation is needed. [4]

Think: How many routes can the charge take in each case?

Series: the charge must pass through every component in the single loop, so an extra resistor adds extra opposition (2). Parallel: an extra resistor adds an extra route for charge, so more current flows in total for the same supply pd — more current at the same pd means the total resistance has gone down (2).

6) **Explain** why the lights in a house are wired in parallel rather than in series. Refer to potential difference in your answer. [4]

In parallel every lamp gets the full supply pd, so each works at full brightness (1–2). Each lamp can be switched on and off independently, because each is on its own loop (1). If one lamp breaks, the others stay on — in series a single failure breaks the only loop and everything goes out (1).

Total: 21 marks. Due Friday 11 September — bring it to the lesson. If you are stuck, start from $V = IR$ and the rules table on the circuits handout.